

Crownstone Hotels UK - Comprehensive Analytical Report

Dataset coverage: 2024-01-01 to 2026-09-05. Six UK hotels with linked commercial, guest, payment, review, workforce and operating-cost data.

Metric	Result	Meaning
Bookings	5,000	Core reservation records
Checked-out stays	4,286	85.7% of bookings
Cancellation rate	11.2%	Cancelled reservations / total
Booking value	£2,254,262	Recognised booking value
Net revenue after commission	£2,108,725	Revenue after distribution commission
Channel commission	£145,538	Distribution cost
ADR	£130.91	Net room revenue / checked-out room nights
Average stay	2.55 nights	Average booking stay length
Average lead time	22.8 days	Booking-to-arrival interval
Average review score	4.19/5	Overall guest rating
Operating cost variance	£389,652 (2.0%)	Actual less budget
Contribution proxy	£-18,222,569	Net revenue after commission less modelled operating costs

1. Data Structure and Cleaning

The workbook is a relational hotel analytics model. Hotels, RoomTypes, Channels and Guests are dimensions; Bookings is the main commercial fact table; Payments, Reviews, Staff and Operating_Costs provide finance, experience, HR and cost facts.

- Converted Excel serial date fields to ISO dates in the cleaned exports.
- Kept legitimate transactional blanks as nulls rather than filling them with artificial values.
- Cleaned Legacy_OTA_Import by trimming whitespace, standardising source/status/city/room-category labels, converting mixed dates, converting currency text to numeric GBP and standardising country labels.
- Created Bookings_Enriched_Cleaned.csv by joining hotel, channel, room-type and guest attributes and added LeadTimeDays and ADR_GBP.
- Kept legacy OTA rows as a governed staging table because they do not contain the full canonical foreign keys required for safe append and deduplication.

2. Commercial Performance

The chain has 5,000 core bookings and £2,254,262 of booking value. After £145,538 of channel commission, net revenue is £2,108,725. Crownstone Regent Quay is the strongest property by net revenue at £518,617; the strongest channel group is Direct, and Booking.com contributes the largest booking volume.

ADR is £130.91, average booking value is £450.85, average stay is 2.55 nights and average lead time is 22.8 days. Expedia has the highest source cancellation rate at 12.3%.

3. Guest and Loyalty

The guest table contains 1,800 records. Among booked guests, 68.4% have more than one booking. Average review score is 4.19/5 and management response rate is 84.6%. Crownstone Calton View has the highest average hotel score at 4.35/5, while Crownstone Temple Row is lowest at 4.00/5.

Rating dimensions: Location 4.25/5, Cleanliness 4.23/5, Service 4.15/5, Value 4.07/5.

4. Operations and Cost

Operating budget is £19,941,642 versus £20,331,294 actual, producing £389,652 variance (2.0%). The largest actual category is Payroll & Agency Staff at £7,807,323; the largest aggregate overrun is Payroll & Agency Staff at £79,540. Net revenue after commission less modelled operating costs gives a contribution proxy of £-18,222,569 (-864.2%).

5. Workforce

There are 252 staff records and 234 active employees. Active annualised salary is approximately £6,981,940. Average performance is 3.85/5, average training is 20.6 hours YTD, and average absence is 7.8 days YTD.

6. Recommendations and Reasons

Increase profitable direct bookings

Reason: channel commission totals £145,538.

Action: use member-only rates, loyalty benefits, direct rebooking and CRM remarketing to reduce commission leakage.

Reduce source-specific cancellations

Reason: Expedia has the highest source cancellation rate at 12.3%.

Action: review flexible-rate exposure, deposit/pre-authorisation rules and cancellation windows by source.

Improve the weakest guest-rating dimension

Reason: the lowest chain-wide dimension is Value at 4.07/5.

Action: create hotel-level service improvement plans and track monthly review-score movement.

Control recurring cost overruns

Reason: Payroll & Agency Staff has the largest aggregate overrun at £79,540.

Action: assign budget owners, threshold alerts and corrective actions for persistent positive variance.

Manage hotels on contribution, not revenue alone

Reason: net revenue after commission is £2,108,725 while modelled operating costs are £20,331,294.

Action: use a hotel scorecard combining revenue, commission, costs, reviews and workforce measures.

Use repeat and loyalty demand to lower acquisition cost

Reason: 68.4% of booked guests are repeat guests.

Action: segment repeat/high-value guests and target direct pre-arrival and rebooking campaigns.

Link workforce measures to service outcomes

Reason: training averages 20.6 hours and absence averages 7.8 days YTD.

Action: compare departmental training, absence and performance with service review scores and hotel KPIs.

Govern legacy OTA integration

Reason: 500 legacy rows were cleaned, but canonical relational keys are incomplete.

Action: map hotel/channel/guest/room keys and run duplicate checks before appending into the production booking fact.

7. Dashboard Design

The Streamlit application uses a pastel-and-white theme, list-style sidebar slicers and separate KPI-card sets for Executive Overview, Revenue & Channels, Operations & Cost, Guest Experience, Workforce, and Booking Funnel & Insights. All Plotly charts and KPI cards recalculate from active filters; related payment/review/staff/cost tables are filtered through booking or hotel keys.

The uploaded workbook is not a recruitment dataset, so a recruitment funnel would be invalid. The dashboard instead uses the supported booking lifecycle funnel: Reservations -> Not Cancelled -> Checked Out -> Reviewed.

8. Data Quality Notes

Table	Rows	Null cell %
Hotels	6	0.00%
RoomTypes	36	0.00%

Table	Rows	Null cell %
Channels	7	0.00%
Guests	1,800	3.17%
Bookings	5,000	9.95%
Payments	5,006	0.00%
Reviews	1,316	0.00%
Staff	252	0.00%
Operating_Costs	1,728	0.00%

Null values are not automatically data errors. Cancellation fields are expected to be blank for non-cancelled bookings, corporate-account data is conditional, and member-since is not applicable to non-members.